

Planning models cont.

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1 Problem 1

A company produces two products sharing the same production capacity.

The production manager plans production based on a forecast of how much is going to be shipped out during a given horizon.

Some production capacity is lost due to setup times (changeover times) if both products are produced in the same week.

The production manager tries to produce only one product per week, but often needs to produce both products.

Products are stored in inventory silos. Each product has a dedicated inventory capacity

Demand must be covered, and production and inventory capacities must be respected

Build a mathematical planning model. The objective function should be to maximize the total value of production, minus inventory costs.

Data:

Current forecast:

	week 1	week 2	week 3	week 4	week 5	week 6	week 7	week 8	week 9
Product 1	6500	5900	6500	7700	8300	7700	5800	7700	8600
Product 2	4800	4700	5900	8200	16,400	5000	13,000	12,500	11,000

Production capacity is 20,000 tons per week if one product is produced during the week. If both products are produced, the total capacity is reduced to 16,000 tons

	value per ton	inventory cost /t/w	init inventory	inventory cap.
Product 1	240	8	3700	9000
Product 2	285	9	6500	10,500

Sets

- $P = \{1, 2\}$: products
- $T = \{1, \dots, 9\}$: set of weeks

Parameters

- $d_{p,t}$: demand for product p at time t
- v_p : value per ton of product p
- h_p : inventory cost per ton per week for product p
- I_p^0 : initial inventory of product p
- $\bar{I}_p$: inventory capacity for product p

Decision variables

- $x_{p,t} \geq 0$: tons of product p produced in week t
- $I_{p,t} \geq 0$: inventory of product p at the end of week t
- $y_{p,t} \in \{0, 1\}$: 1 if product p is produced in week t , 0 otherwise
- $z_t \in \{0, 1\}$: 1 if both products are produced in week t , 0 otherwise

Model

$$\begin{aligned} \min Z &= \sum_{p \in P} \sum_{t \in T} v_p x_{p,t} - \sum_{p \in P} \sum_{t \in T} h_p I_{p,t} \\ \text{s.t.} \\ I_{p,t} &= I_{p,t-1} + x_{p,t} - d_{p,t} \quad \forall p \in P, t \in T, I_{p,0} = I_p^0 \\ I_{p,t} &\leq \bar{I}_p \quad \forall p \in P, t \in T \\ x_{p,t} &\leq C^{\text{single}} y_{p,t} \quad \forall p \in P, t \in T \\ z_t &\geq y_{1,t} + y_{2,t} - 1 \quad \forall t \in T \\ \sum_{p \in P} &\leq 20,000 - 4,000 z_t \quad \forall t \in T \end{aligned}$$

2 Static planning model

Basic model: the economic order quantity model (EOQ)

- Assumptions:
 - Demand is constant and continuous – D per year
 - The lot size (decision variable) is Q
 - A fixed cost (order cost or setup cost) – A – is incurred every time a replenishment (order) takes place
- Example:
 - $D = 1000$ per year
 - $A = 200$ per replenishment If the lot size $Q = 100$, the number of replenishments per year = $D/Q = 1000/100 = 10$
Total order costs per year = $10 \times 200 = 2,000$
- To minimize these fixed costs: maximize Q , to get as few replenishments as possible
Inventory cost is assumed to be an annual inventory rate, r , multiplied with the value of the product, v
Assume $r = 0.1$ and $v = 500$
Inventory holding cost per year for $Q = 100$:

$$v \times r \times Q/2 = 500 \times 0.1 \times 100/2 = 2500$$

(remember: $100/2 = 50$ is the average inventory for $Q = 100$)

3 Finding the optimal trade-off

between order costs and inventory holding costs

Total relevant costs (TRC): total order costs + total inventory holding costs

$$= A \times D/Q + v \times r \times Q/2$$

In the example above, we assume $Q = 100$, $r = 0.1$ and $v = 500$, $\text{TRC} =$

$$\begin{aligned} & 200 \times 1000/100 + 500 \times 0.1 \times 100/2 \\ & = 2000 + 2500 = 4500 \end{aligned}$$

But which level of Q will minimize TCR?

$$\begin{aligned} \text{TCR}'_Q &= -\frac{AD}{Q^2} + \frac{vr}{2} \rightarrow -\frac{AD}{Q^2} + \frac{vr}{2} = 0 \\ Q &= \sqrt{\frac{2AD}{vr}} \end{aligned}$$

(this is the “famous” EOQ formula)

4 The EOQ formula with quantity discounts

Data:

- Demand per year $D = 1000$
- Ordering cost $A = 200$
- Value per unit $v = 500$
- Interest rate per year $r = 0.1$

Assume the cost price will be 499 instead of 500 if the order quantity is at least 150 units (the breakpoint). Remember, the previous EOQ was 84.44

- Let $v_0 = 500$ be the price without the quantity discount
- and $v_0 \times (1 - d) = 499$ be the price with quantity discount
- hence, d is the percentage discount, in this case $(500 - 499)/500 = 0.02$

The TRC when there are quantity discounts

- The purchasing price v must now be included in the TRC
- Without discount: $\text{TRC} = AD/Q + Qv_0r/2 + Dv_0$
- With discount: $\text{TRC} = AD/Q + Qv_0(1 - d)r/2 + Dv_0(1 - d)$
(but only for $Q \geq Q_b$ where b is the breakpoint)